HOMEWORK 1 2026

1)

To address the first question, we must first determine the input power (P_{in}) delivered by the generator with an 50Ω load. Assuming the generator's displayed amplitude of $0.1V$ refers to the peak-to-peak voltage in an open-circuit condition ($V_{pp,oc} = 0.1V$) the peak voltage is:

$$V_p = \frac{0.1V}{2 \cdot 2} = 0.025V$$

(Using different assumptions for the amplitude results in an unjustifiable IL).

The root-mean-square (RMS) voltage is therefore:

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$

The input power, expressed in Watts and subsequently converted to dBm, is computed as follows:

P_in(W) = V_rms^2 / R = (0.025)^2 / (2 * 50) = 6.25 μW

$$P_{in}(dBm) = 10 \cdot \log_{10}(P_{in}(mW)) \approx -22dBm$$

Once P_{in} is calculated, the total Insertion Loss (IL) can be derived using the measured output power ($P_{out} = -30dBm$):

$$IL(dB) = P_{in}(dBm) - P_{out}(dBm) = -22dBm - (-30dBm) = \textbf{8dB}$$

This Insertion Loss result is physically consistent with the experimental setup. The ideal combiner is expected to introduce an insertion loss of $6 - 7dB$. The remaining $1 - 2dB$ can be reasonably attributed to mismatch losses (due to the un-terminated port of the combiner) and minor connector dissipations. The coaxial cables introduce a negligible attenuation at this frequency (approximately $3dB$ per $100m$ at $11MHz$), which safely justifies ignoring their contribution to the total loss in our short connections

2)

To analyze the measurement results in the table, we first define the fixed parameters and offset terms:

$$P = (9 + 9 + 3) / 3 = 7 \text{ dB}$$
$$G_{att} = -29.5 \text{ dB} \quad \text{(Attenuator Gain)}$$

Using these parameters, the remaining columns of the table are calculated using the following relationships:

$$P_{out,amp} = P_{out} + P - G_{att}$$
$$P_{in,amp} = P_{in} - IL$$
$$G_{amp} = P_{out,amp} - P_{in,amp}$$

A (V)	P_out (dBm)	P_in,amp (dBm)	P_out,amp (dBm)	G_amp (dB)
0,1	−38	−30,0	−1,5	28,5
0,15	−35	−26,5	1,5	28,0
0,2	−32,5	−24,0	4,0	28,0
0,3	−28,4	−20,5	8,1	28,6
0,4	−26,4	−18,0	10,1	28,1
0,5	−24	−16,0	12,5	28,5
0,6	−22,2	−14,4	14,3	28,7
0,8	−19,7	−11,9	16,8	28,7
0,9	−18,7	−10,9	17,8	28,7
1,0	−17,6	−10,0	18,9	28,9
1,2	−16,2	−8,4	20,3	28,7
1,4	−15,2	−7,1	21,3	28,4

In conclusion, the overall amplifier gain (*G_amp*) can be determined by averaging the calculated *G_amp* values across all measurements in the table, resulting in an average gain of **28.5 dB**.

3)

Frequency	P_out(dBm)	P_out + P (dBm)
f1 = 10 MHz	-18	-11

Frequency	P_out(dBm)	P_out + P (dBm)
f2 = 11 MHz	-18	-11
2f1-f2 = 9 MHz (IM3)	-48	-41
2f2-f1 = 12 MHz (IM3)	-47	-40

First, we calculate the average power of the fundamental frequencies, P_out = -11 dBm, and the average power of the third-order intermodulation products, P_IM3 = -40.5 dBm.

Using these values, we determine the difference between the output power of the fundamental frequencies and the IM3 products:

$$\Delta IM = -11 - (-40{,}5) = \mathbf{29{,}5\ dB}.$$

Next, we calculate the output power per tone at the amplifier output by accounting for the 29.5 dB attenuator:

$$P_{fund,out} = -11 + 29.5 = \mathbf{18.5\ dBm.}$$

Since the IM3 products increase at a rate of 3 dB for every 1 dB increase in the fundamental signal, the output third-order intercept point (OIP3) is calculated as follows:

$$OIP3 = P_{fund,out} + \Delta IM/2 = 18.5 + 29.5/2 \approx \mathbf{33.3\ dBm.}$$

Finally, the input third-order intercept point (IIP3) is derived using the gain G = 28.5 dB obtained in question 2:

$$IIP3 = OIP3 - G = 33.3 - 28.5 \approx \mathbf{4.8\ dBm.}$$

4)

The input power of the PA is Pi= -21.4 + 28.5 - 10=-2.9dBM. A gain of G=22.6+2.9=25.5dBm is needed for the amplifier to work correctly. The **ZVA-183W-S+** serves perfectly this purpose: has a low power consumption, 625 mA and 15V. Ensures a good margin against the 25.5 dBm, since it has a G of 28dBm. Has a CI = 2*(34.5-21.6)+6 = 31.8 dBm that fits the requirements. It costs $2715.

5)

We must first calculate the contributions of each block of the system (using its gain and noise figure).

Starting from the DUT:
$G_{TOT} = 28.5\,dB = 707.5$ which is a value given by the previous points' resolution, and $NF_{DUT} = 4\,dB$.

We calculate the temperature contribution using

$$T_{DUT} = (NF - 1)\,T_0 = (10^{\frac{4}{10}} - 1)(290\,K) = 438.5\,K$$

For the first attenuator:

$G_{att1} = 10^{\frac{-10}{10}} = 0.1$ and the temperature contribution is

$$T_{att1} = (NF - 1)\,T_0 = (10^{\frac{10}{10}} - 1)(290\,K) = 2610\,K$$

For the PA amplifier:

$G_{PA} = 29\,dB = 794.3$ and $NF_{PA} = 3\,dB$ which is given by its datasheet, based on the PA chosen on the previous question. The temperature contribution is then calculated using

$$T_{PA} = (NF - 1)\,T_0 = (10^{\frac{3}{10}} - 1)(290\,K) = 288.6\,K$$

For the second attenuator:

$G_{att1} = 10^{\frac{-1.4}{10}} = 0.7244$ and the temperature contribution is

$$T_{att1} = (NF - 1)\,T_0 = (10^{\frac{1.4}{10}} - 1)(290\,K) = 110.3\,K$$

Then, we must calculate the equivalent temperature contribution by the cascade of blocks plus considering the one from the receiving antenna, as follows:

$$T_{eq} = T_{DUT} + \frac{T_{att1}}{G_{DUT}} + \frac{T_{PA}}{G_{DUT}G_{att1}} + \frac{T_{att2}}{G_{DUT}G_{att1}G_{PA}} + T_A = 438.5\,K + \frac{2610\,K}{707.5} + \frac{288.6\,K}{707.5\,(0.1)} + \frac{110.3\,K}{707.5\,(0.1)\,(794.3)}$$

$$+ 120\,K = 566.3\,K$$

Now we have to calculate the noise power that goes into the antenna, using $P_{in} = k_B T_{eq} BW = 1.38 \times 10^{-23}\frac{J}{K}(566.3\,K)\,(2\,MHz) = 1.56 \times 10^{-14} W$ and we can now calculate the noise power that goes out from the antenna:

$P_{out} = P_{in} G_{tot} = 1.56 \times 10^{-14} W(10^{28.5-10+29-1.4}) = 6.35 \times 10^{-10} W$ which converted to dBm corresponds to $-61.97\,dBm$.

The noise power at the retransmitting antenna, within a 2 MHz bandwidth, is estimated to be approximately **−62.0 dBm**, assuming linear operation and the typical component specifications used in the calculations.

The frontend can suffer from intermodulation distortion, especially third-order intermodulation products generated by the power amplifier when amplifying the two-tone signal. These products can interfere with the desired signal and degrade the carrier-to-intermodulation ratio. The PA may also experience gain compression or saturation if operated beyond its linear range. Furthermore, the noise generated by the components reduces the SNR, while inadequate amplifier stability can cause unwanted oscillations.

Lorenzo Bigucci 10917917 (334649)

Isaac Santamaria 10926289 (336643)

Edoardo Marziali 240539